BUILDER SKILLS

From spec to proven delivery.

AI agents that design, build and test Itential automation, with an approval at every stage.

19 skills | **23** built-in specs | **6** stages · **4** agents | Claude Code · Codex · Copilot · Cursor

WHAT YOU GET

Evidence at every stage
A spec, a design, test results and an as-built record, not just a working job

The right execution model
Workflow, LCM, Gateway5 service or agent, chosen per phase with a stated reason

Any AI tool
The same skills in Claude Code, Codex, Copilot and Cursor

Your rules, kept
Org, team and personal standards that updates never overwrite

WHY SPEC-DRIVEN

Delivery needs an operating model.

WITHOUT ONE

- **Intent stays implicit** — "done" lives in someone's head
- **Discovery happens late** — the missing adapter shows up mid-build
- **Design and build blur** — decisions made inside the canvas
- **"It works" is never proven** — a green job isn't evidence
- **Knowledge walks away** — the next team starts from zero

WITH SPEC-DRIVEN DEVELOPMENT

Lock intent before touching the environment.
Prove the result before calling it done.

Requirements → Feasibility → Design → Build → Test → As-Built

Each stage: one artifact, one approval. Nothing moves forward on an assumption.

DELIVERY LIFECYCLE

Six stages. Four agents. Full traceability.

01 APPROVE

Requirements

Spec Agent

`customer-spec.md`

02 APPROVE

Feasibility

Solution Arch. Agent

`feasibility.md`

03 APPROVE

Design

Solution Arch. Agent

`solution-design.md`

04 HAND OFF

Build

Builder Agent

`project + assets`

05 APPROVE PLAN

Test

QA Agent

`test-plan · test-report`

06 SIGN OFF

As-Built

QA Agent

`as-built.md`

1 · REQUIREMENTS

- Start from 23 built-in specs or a blank one
- Capture business context, scope and constraints
- Write the acceptance criteria
- Engineer approves before anything else happens

CUSTOMER RECEIVES

An approved high-level design: what will be built, scope, constraints and how it will be accepted.

2 · FEASIBILITY

- First contact with the platform: authenticate and pull live data
- Check adapters, integrations, and what LCM, agents and Gateway5 already offer
- Find existing assets to reuse
- Decide: feasible, feasible with constraints, or not

CUSTOMER RECEIVES

A feasibility decision with constraints flagged and reuse opportunities named.

3 · DESIGN

- Component inventory: build or reuse
- An execution model for each phase, with the reason, confirmed with the engineer
- Adapter mappings and build order
- Each acceptance criterion mapped to a test

CUSTOMER RECEIVES

A low-level design the builder can follow and QA can test against.

4 · BUILD

- Children first, orchestrator last, each one tested
- runCode + Enable Query instead of long utility chains
- Deep validation before every save; zero errors and warnings
- Atomic project import, owners set right after, and a question before replacing an existing project

CUSTOMER RECEIVES

A delivered project with every component tested and real IDs recorded, ready for acceptance testing.

5 · TEST

- Test plan drafted from the acceptance criteria
- Engineer approves the test data before anything runs live
- Static and live acceptance cases, including agent sessions, with evidence
- Failures go back to the builder; QA never edits workflows

CUSTOMER RECEIVES

Pass or fail with evidence for every acceptance criterion; pending items called out, not hidden.

6 · AS-BUILT

- Written once every case passes, or residuals are accepted
- Deviations recorded in the design
- Scope changes dated in the spec
- Engineer signs off

CUSTOMER RECEIVES

The delivered state, deviations and learnings, backed by real test evidence. The baseline for the next change.

DESIGN DECISION

Pick how each phase runs.

WORKFLOW

The default

Same steps, same result, every run. Used unless another model has a specific reason in this use case.

Repeatable changes, checks and orchestration

LIFECYCLE MANAGER

A thing that outlives the run

Stores the instance and runs day-2 actions against it: a port service, a circuit, a network.

Not for one-off tasks such as a backup or a report

GATEWAY5 SERVICE

Code that runs on a gateway

Python, Ansible or OpenTofu defined in YAML and called from a workflow.

When the work is a script, playbook or plan

FLOWAI AGENT

Judgment at runtime

An agent reasons with platform tools. A parent can call child agents; a child is judged by its answer, not its status.

When the path can't be written down in advance

How the choice is made

Feasibility records what the platform already has. Design asks only the questions the spec leaves open, proposes a model for each phase with the reason, and the engineer confirms it.

Agent first, then deterministic

Build and test an agent to find out what the process should be, then turn it into a deterministic workflow spec. Or build the agent as the delivered answer, or go straight to workflows.

19 SKILLS

Start wherever you are.

Deliver from a spec

4

The lifecycle, stage by stage

/spec-agent

Use it to start a new delivery. Refines the use case, sets scope and acceptance criteria, and produces the approved customer-spec.md — from 23 built-in specs or from scratch.

/solution-arch-agent

Use it once requirements are approved. Connects to the platform, assesses feasibility, then writes the design: components, execution model per phase, adapter mappings, build order and tests.

/builder-agent

Use it once the design is approved. Builds and tests every component — projects, workflows, templates, MOP, agents — children first, orchestrator last.

/qa-agent

Use it after the build. Acceptance tests against the approved criteria with real data, then the as-built record. Failures go back to the builder.

Start from what exists

4

Turn running things into specs

/flowagent-to-spec

Use it when a FlowAI agent proved the pattern. Reads the agent and its sessions, including child agents it calls, and writes a deterministic workflow spec.

/project-to-spec

Use it when automation works but isn't documented. Reads a project and writes customer-spec.md and solution-design.md.

/documentation

Use it to document a whole platform. Surveys global assets, groups them by use case and writes a design for each.

/explore

Use it to look around first. Authenticate, pull live platform data and browse — no lifecycle required.

ITENTIAL

Build on the platform

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Deep knowledge per area

/flowagent

FlowAI agents: LLM profiles, tools, decorators, human approvals (ViewData, InteractiveHTML, QuickForm), agents that call other agents, attached skills, export and import.

/flowagent-skill-builder

Write, package, import and test skills for FlowAI agents, in the Agent Skills format.

/iag

Gateway5 services: Python, Ansible and OpenTofu defined in YAML and called from workflows with runService.

/itential-mop

Command templates with validation rules for pre- and post-checks.

/itential-devices

Devices, backups, diffs and device groups.

/itential-golden-config

Golden config trees, compliance plans and grading. Detects drift; never applies remediation.

/itential-inventory

Device inventories, nodes, actions, tags and broker-action inventories.

/itential-lcm

Lifecycle Manager: resource models, instances and lifecycle actions.

/itential-json-forms

JSON Forms for manual triggers and tasks: static, REST-bound and cascading dropdowns.

Modernize

2

Move estates forward

/gateway4-to-gateway5

Use it before a Gateway4 → Gateway5 move. Read-only readiness report and manual-action guide; the migration itself then runs through /spec-agent with the migration spec and converter.

/jst-to-code

Use it to retire transformations. Converts JSTs one at a time into single-file Python for the Run Code task, each proven against test data from your samples or a past job. Read-only on the platform.

GUARDRAILS

Built into every stage, not bolted on.

APPROVAL GATES

The engineer decides

The spec, feasibility, design, test plan and as-built record each need an explicit approval before the next stage starts.

VALIDATE FIRST

Zero errors, zero warnings

Every workflow is deeply validated before it is saved. A warning blocks delivery the same way an error does.

EVIDENCE

A green job isn't proof

Each task's own output is checked for embedded errors and empty values, and the values are stated before a run is called successful.

SAFE IMPORTS

Ask before replacing

Importing over an existing project replaces it. The builder compares the file with the live project and asks before anything is deleted.

NO DEVICE REMEDIATION

Detect, report, push normally

Golden config finds and reports drift. Corrections go through a normal config-push delivery, never an automatic remediation task.

PERMISSIONS

The platform stays in charge

Skills act with the configured credentials and never grant themselves access. Customization can narrow the rules but can't weaken a safety rule.

GET STARTED

Install it. Make it yours. Pick a spec.

Install in your AI tool

Claude Code	`/plugin marketplace add itential/builder-skills` `/plugin install itential-builder@itential-builder`
Codex CLI	`codex plugin marketplace add itential/builder-skills` `codex plugin add itential-builder@itential-builder`
Copilot · VS Code	`Chat: Install Plugin From Source → itential/builder-skills`
Copilot CLI	`copilot plugin marketplace add itential/builder-skills` `copilot plugin install itential-builder@itential-builder`
Cursor	`gh skill install itential/builder-skills --agent cursor --all`

Check, run and update steps for each tool: `docs/vendor-install.md`. Watch releases on the repo to hear about updates.

Customize without forking

Each skill has `custom/org`, `custom/team` and `custom/dev` for your standards. Most specific wins; updates never overwrite them; no layer can weaken a safety rule.

23 built-in specs

Port turn-up · VLAN · BGP peer · VPN tunnel · Circuit · WAN bandwidth · Load balancer VIP · Device onboarding · Decommissioning · Software upgrade · Change management · Health check · Config backup · Drift remediation · Compliance audit · Incident remediation · Firewall rules · Cloud security groups · SSL certificates · IPAM · DNS records · AWS web server · Gateway4 → 5

Three packs: **Builder** delivers and tests · **Operator** runs delivered automations · **Admin** keeps the platform healthy

HOW IT RUNS

A delivery in four steps.

```
› spec-agent — pick a use case, lock requirements
  customer-spec.md approved
› solution-arch-agent — check the platform, design it
  feasibility.md + solution-design.md approved
› builder-agent — build and test each component
  project delivered, handed to QA
› qa-agent — test plan, live acceptance tests, as-built
  test-report.md + as-built.md signed off
```

Invoke a skill as `/itential-builder:spec-agent` in Claude Code, `$itential-builder:spec-agent` in Codex, or `/spec-agent` in Copilot and Cursor.